Project Title: Construction of Radar Using Arduino

Submitted by

ARUN KUMAR (22116109048)

ROHIT KUMAR(22116109005)

RITIK KUMAR(22116109010)

PRIYANSHU KUMAR

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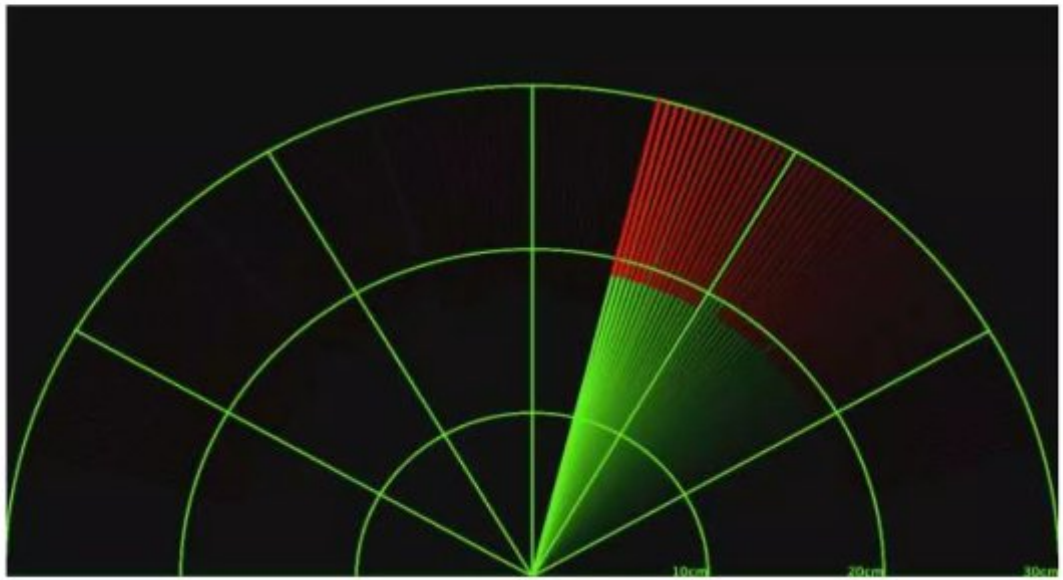
Introduction

- Design a simple Radar application using Arduino and processing IDE software. This Arduino radar is implemented with the help of processing application.
- Radar is a long-range object detection system that uses radio waves to establish certain parameters of an object such like its range, speed and position.
- The project is based on the Sonar technology and will be using an ultrasonic sensor to determine the presence of any object in a particular range.



Literature Review Summary

S. No.	Paper & Publication	Author	Content
1.	Arduino Based Radar System	Anuj Dutt, 2014	Basic Introduction of Radar and Components requirement
2.	Ultrasonic Radar	Nithya V, Robin Alias, Shreepriya N.S, Vivek Joseph, Yathukrishna K.T.	Practical Implementation and Experimental Results of Ultrasonic Radar
3.	Implementation of Radar using Ultrasonic Sensor and Arduino	Srijan Dubey, Supragya Tiwari, Sumit Roy	Study of Implementation and collecting outputs of Radar using Arduino

Components Required

Some Hardware and some Software are used in constructing the Radar. List of Hardware and software used are as follows-

- Hardware-

  1- Computer for writing the programming codes
  2- Arduino UNO Board
  3- Bluetooth Cable (connecting Arduino with Computer)
  4- Ultrasonic Sensor (HC-SR04)
  5- Servo Motor (SG90)
  6- Connecting & jump wires
  7- Breadboard

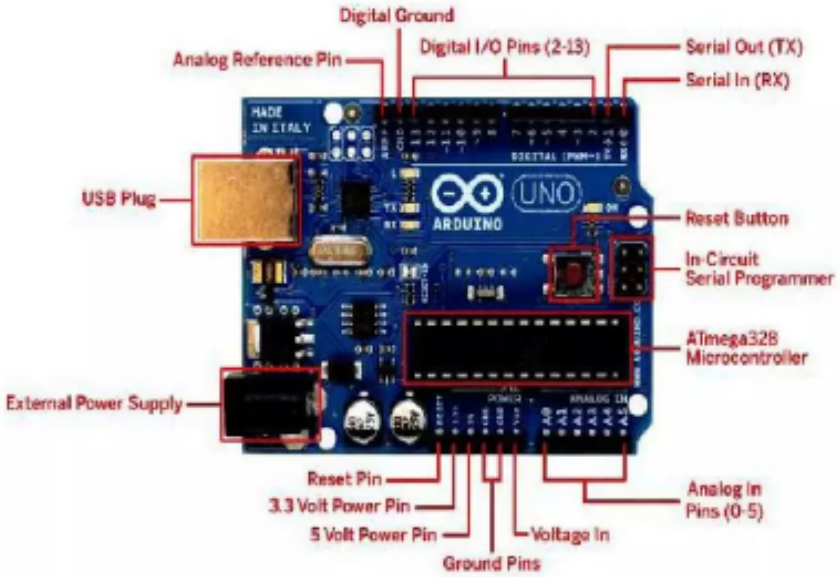
- Software-

  1- Arduino IDE
  2- Processing IDE / MATLAB

Arduino UNO Board

The Arduino UNO board is a microcontroller based on the AT mega328. It has 14 digital Input / Output pins in which 6 can be used as PWM outputs, a 16 MHZ ceramic resonator, an ICSP header, a USB connection, 6 analog inputs, a power jack and a reset button. This contains all the required support needed for microcontroller.

Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment) that runs on our computer, used to write and upload computer code to the physical board.

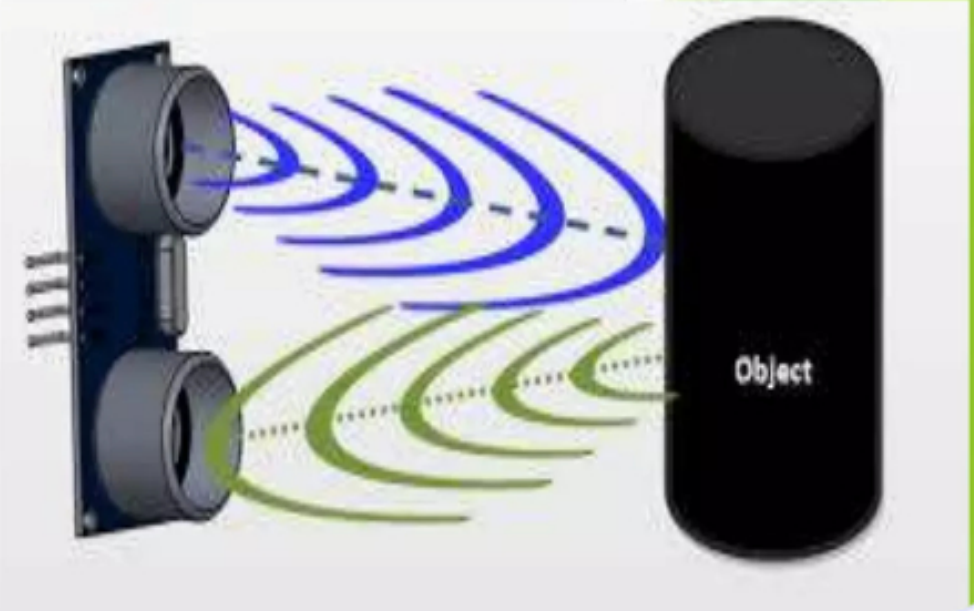


Features of Arduino UNO	
Microcontroller	AT mega 328
Input Voltage (Recommended)	7-12V
Operating Voltage	5V
Input Voltage (Limits)	6-20V
Digital I/O pins	14(6 of them provide PWM Output)
Analog Input Pins	6
DC current per I/O Pin	40ma
DC current for 3.3V Pin	50ma
Flash Memory	32KB (AT mega 328) 0.5 KB used by Bootloader
SRAM	2KB (AT mega 328)
EEPROM	1KB (AT mega 328)
Clock Speed	16 MHZ

Ultrasonic Sensor (HC-SR04)

The **HC-SR04** ultrasonic sensor uses sonar to determine distance to an object like bats or dolphins do. It offers excellent non-contact range detection with high accuracy and stable readings in an easy-to-use package. From 2cm to 400 cm or 1" to 13 feet.

It emits an **ultrasound** at 40 000 Hz which travels through the air and if there is an object or obstacle on its path It will bounce back to the module. Considering the travel time and the speed of the sound you can calculate the distance. The **HC-SR04 Ultrasonic** module has 4 pins, **Ground**, **VCC**, **Trig** and **Echo**.



Servo Motor (SG-90)

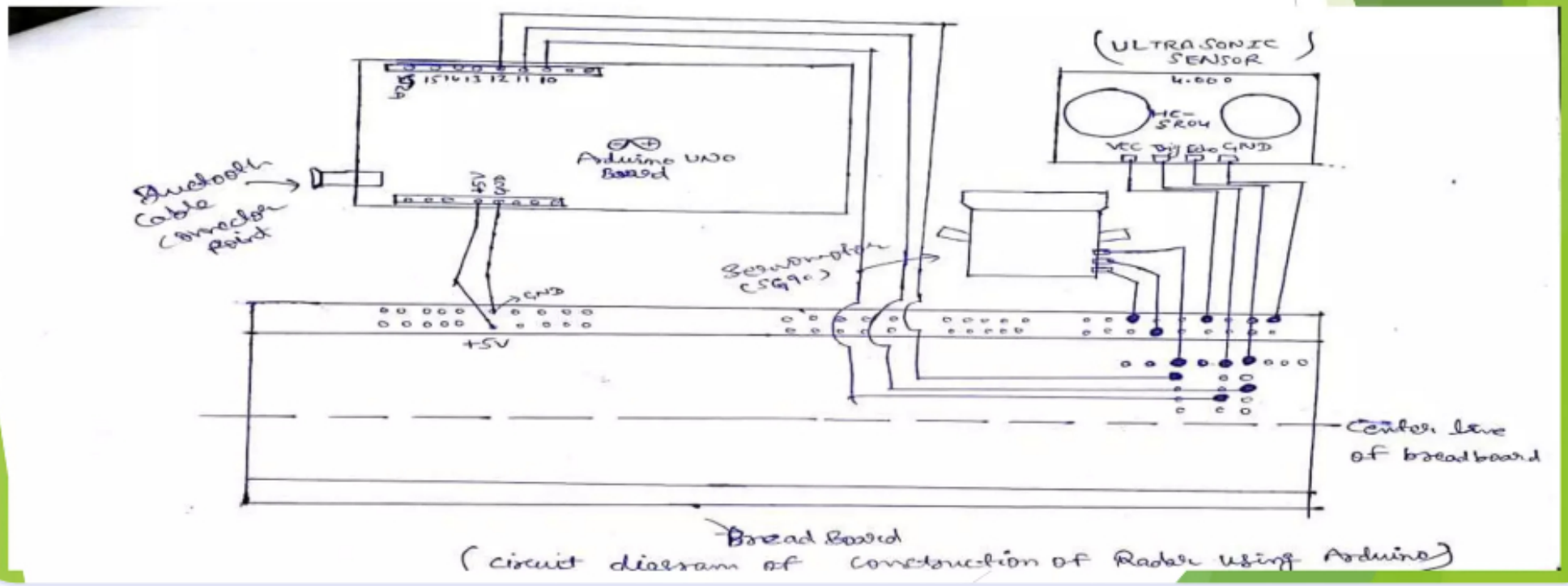
servo motors can rotate from 0° to 180° due to their gear arrangement. The gears in the motors are easily subjected to wear and tear, so if our application requires stronger and long running motors we can go with metal gears or just stick with normal plastic gear.

the most important parameter, which is the **torque** at which the motor operates. Tower Pro SG-90 motor provides the **Torque** of 2.5 Kg/cm. There are three wires in the servo motor. **Red** one indicates the **+5V**, **Brown** indicates **grounding** and **Orange** wire shows the **PWM.**

Working & Circuit Diagram

Initially code should be upload to the Arduino after making the connection. We can observe the servo motor sweeping from 0° to 180^0 and again back to 0°.

Since the ultrasonic sensor is mounted over the servo motor so it will also participate in the sweeping action. The graphical representation of the data from the ultrasonic sensor in a radar type display. If the ultrasonic sensor detects any object within its range the graphical representation will be shown on the screen.



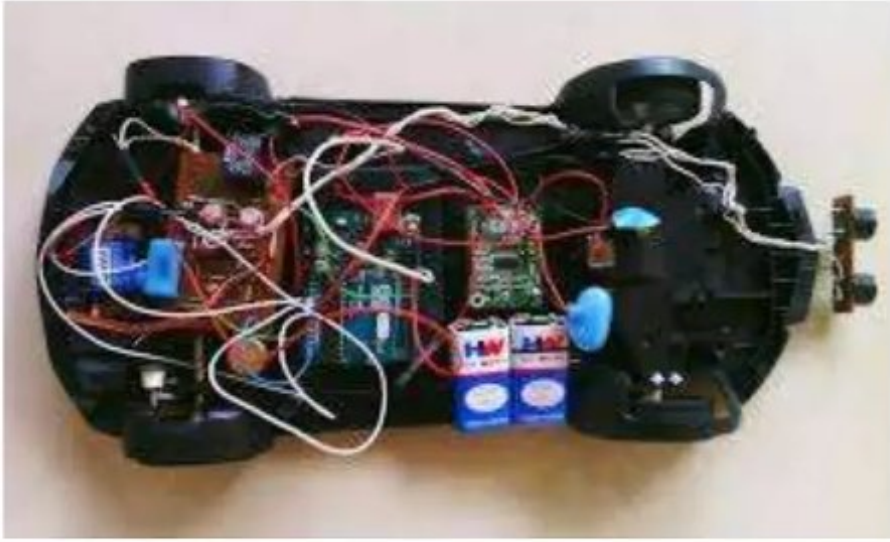
(circuit diagram of construction of Radar using Arduino)

Objective

- The objective of the project is to detect the obstacle using Ultrasonic Sensor, Arduino UNO board and MATLAB or Processing coding as a platform to display the results.
- To prevent or reduce the accidents and useful for security and protection.
- Detection of objects come in the range of 2 to 3 meters and give signal to the observer about the object.

Applications

- In underwater networks.
- In driverless vehicle system.
- In speed detection of mobile objects.
- In various military operations such as to guide automatic weapons.
- Important part of the air defense system, operation of offensive missiles and other weapons.
- For navigation, ship security, remote sensing, space (satellites).
- Detect and tracks the target, even control the weapon.

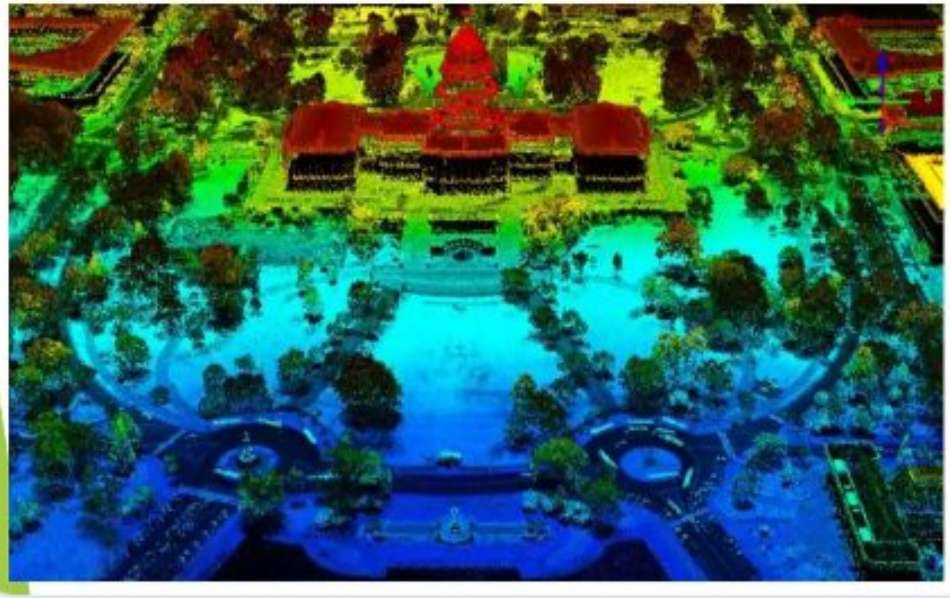


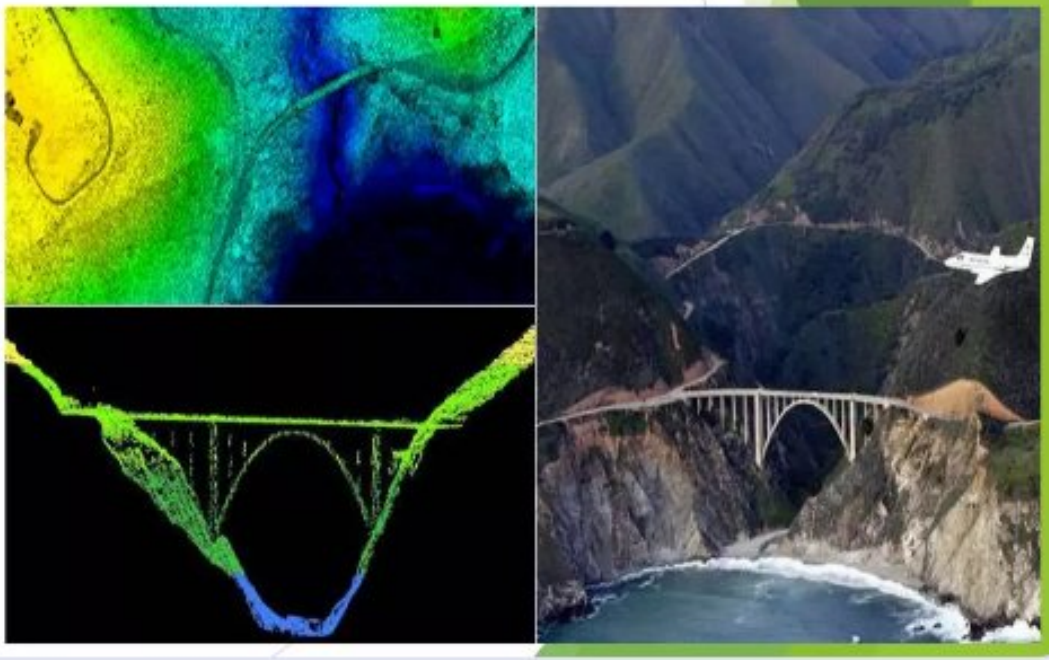
Conclusion & Expected Outcome

- Radar is used to find velocity, long-range and position of the object. Advantage of RADAR is that it provide superior penetration capability through any type of weather condition.
- Range of the object detection for the current project is **between 2 to 3 meters**.
- To increase the range of object detection, Radar can be made more improved which is known as "Lidar". Lidar is advanced type of Radar which uses visible light from laser.

Future Scope

Future scope of Radar is a new and advanced technology “Lidar”. Lidar which stands for Light Detection and Ranging, is a remote sensing method that uses light in the form of pulsed laser to measure ranges (variable distances) to the Earth. LIDAR technology is used to create 3D image of the object that is in distance (3D gated viewing laser radar).





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THANK YOU